

Task 3

R code:

```
# Get current working directory
getwd()

#read dataset

data_for_analysis <- read.csv("/Users/apple/Downloads/INRTU courses/2nd
semester/Data Analysis/Homework 3/data_for_analysis.csv", header=T)

data

#Load required libraries

library(car)

library(lawstat)

library(gtsummary)

library(DataExplorer)


#Data preparation

# Convert outcome to factor
data_for_analysis$outcome <- as.factor(data_for_analysis$outcome)


# Define hormone variables
attach(data_for_analysis)

hormone_vars <- c("hormone1", "hormone2", "hormone3", "hormone4",
                  "hormone5", "hormone6", "hormone7", "hormone8",
                  "hormone10_generated")
```

1. Creating a table with descriptive statistics and specify the parameters according to the selected distribution (by group) (all hormones).

R code:

```
# Create descriptive statistics table for all hormones by outcome group
summary(data_for_analysis)

# First, let's check the automatic table
```

```

tbl_summary(data_for_analysis) # Automatic table for all variables

# Table by groups (by outcome)
tbl_summary(data_for_analysis, by = outcome) # By groups

# Now create a customized table for hormones only
# First, determine distribution type for each hormone
check_distribution <- function(data, var) {
  # Clean data - remove NAs
  clean_data <- data[[var]][!is.na(data[[var]])]

  if(length(clean_data) < 3) {
    return("insufficient data")
  }

  # Shapiro-Wilk test
  shapiro_test <- shapiro.test(clean_data)

  if(shapiro_test$p.value > 0.05) {
    return("normal") # Normally distributed - use mean (SD)
  } else {
    return("non-normal") # Non-normally distributed - use median (IQR)
  }
}

# Check distribution for each hormone
distribution_results <- sapply(hormone_vars, function(h) {
  check_distribution(data_for_analysis, h)
})

```

```

print("Distribution types for hormones:")
print(distribution_results)

# Create customized table for hormones with appropriate statistics by group
# For normally distributed: Mean (SD)
# For non-normally distributed: Median (IQR)

hormone_table_by_group <- data_for_analysis %>%
  select(outcome, all_of(hormone_vars)) %>%
  tbl_summary(
    by = outcome,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})"
    ),
    digits = list(all_continuous() ~ 2)
  ) %>%
  modify_header(label ~ "***Hormone Variable***") %>%
  add_overall() %>%
  bold_labels()

hormone_table_by_group

```

2. Performing the Levene's Test for Homogeneity of Variance and Shapiro-Wilk test

R code:

```

# Create a function to perform both tests
normality_variance_tests <- function(data, hormone_var) {
  # Split data by outcome groups
  group0 <- data[data$outcome == "0", hormone_var]
  group1 <- data[data$outcome == "1", hormone_var]

```

```

# Remove NAs

group0 <- group0[!is.na(group0)]
group1 <- group1[!is.na(group1)]

# Shapiro-Wilk test for each group

shapiro0 <- shapiro.test(group0)
shapiro1 <- shapiro.test(group1)

# Levene's Test

levene_result <- leveneTest(as.formula(paste(hormone_vars, "~ outcome")),
                             data = data)

# Compile results

results <- data.frame(
  Hormone = hormone_var,
  # Group 0 Shapiro-Wilk
  SW_Group0_W = round(shapiro0$statistic, 3),
  SW_Group0_p = round(shapiro0$p.value, 4),
  SW_Group0_Normal = ifelse(shapiro0$p.value > 0.05, "Normal", "Non-
normal"),
  # Group 1 Shapiro-Wilk
  SW_Group1_W = round(shapiro1$statistic, 3),
  SW_Group1_p = round(shapiro1$p.value, 4),
  SW_Group1_Normal = ifelse(shapiro1$p.value > 0.05, "Normal", "Non-
normal"),
  # Levene's Test
  Levene_F = round(levene_result$`F value`[1], 3),
  Levene_p = round(levene_result$`Pr(>F)`[1], 4),
  Levene_Result = ifelse(levene_result$`Pr(>F)`[1] > 0.05,
                          "Equal variances", "Unequal variances")
)

```

```

    return(results)
}

# Apply tests to all hormones
diagnostic_results <- do.call(rbind, lapply(hormone_vars, function(h) {
  normality_variance_tests(data_for_analysis, h)
}))

# Display results
print("QUESTION 2: NORMALITY AND VARIANCE TEST RESULTS")
print(diagnostic_results)

# Create a formatted table for the diagnostic tests
diagnostic_table <- diagnostic_results %>%
  select(Hormone, SW_Group0_Normal, SW_Group1_Normal, Levene_Result)

colnames(diagnostic_table) <- c("Hormone",
                                "Distribution (Group 0)",
                                "Distribution (Group 1)",
                                "Variance Homogeneity")

print("Diagnostic Summary Table:")
print(diagnostic_table)

```

From the output we can see that all 9 hormones show non-normal distribution in both groups (Shapiro-Wilk $p < 0.05$) but have equal variances between groups (Levene's test $p > 0.05$).

3. Q-Q Plots and histograms (all hormones)

R code:

```

for(hormone in hormone_vars) {

```

```
par(mfrow = c(2, 2))

# Extract data for each group
group0_data <- data_for_analysis[data_for_analysis$outcome == "0", hormone]
group1_data <- data_for_analysis[data_for_analysis$outcome == "1", hormone]

# Remove NAs
group0_data <- group0_data[!is.na(group0_data)]
group1_data <- group1_data[!is.na(group1_data)]

# Histogram - Group 0
hist(group0_data, col = "lightblue",
      main = paste(hormone, "- Group 0 Histogram"),
      xlab = "Value")

# Histogram - Group 1
hist(group1_data, col = "lightgreen",
      main = paste(hormone, "- Group 1 Histogram"),
      xlab = "Value")

# Q-Q Plot - Group 0
qqnorm(group0_data,
        main = paste(hormone, "- Group 0 Q-Q Plot"),
        col = "blue", pch = 19)
qqline(group0_data, col = "red", lwd = 2)

# Q-Q Plot - Group 1
qqnorm(group1_data,
        main = paste(hormone, "- Group 1 Q-Q Plot"),
        col = "green", pch = 19)
```

```
qqline(group1_data, col = "red", lwd = 2)

par(mfrow = c(1, 1)) # Reset
}
```

4. Performing the Brunner-Munzel test, t.test, wilcox.test for 2 independent groups (enter the p-value into the data description table) and repeating the process for multiple variables (all hormones)

R code:

```
# (Brunner-Munzel, t-test, Wilcoxon) for all hormones

# Function to perform all three tests
compare_tests <- function(data, hormone_var) {
  # Extract data by group
  group0 <- data[data$outcome == "0", hormone_var]
  group1 <- data[data$outcome == "1", hormone_var]

  # Remove NAs
  group0 <- group0[!is.na(group0)]
  group1 <- group1[!is.na(group1)]

  # t-test
  t_result <- t.test(group0, group1)

  # Wilcoxon test
  wilcox_result <- wilcox.test(group0, group1)

  # Brunner-Munzel test
  bm_result <- brunner.munzel.test(group0, group1)

  # Return p-values
```

```

results <- data.frame(
  Hormone = hormone_var,
  t_test_pvalue = round(t_result$p.value, 4),
  wilcox_pvalue = round(wilcox_result$p.value, 4),
  bm_pvalue = round(bm_result$p.value, 4)
)

return(results)
}

# Apply tests to all hormones
all_comparison_results <- do.call(rbind, lapply(hormone_vars, function(h) {
  compare_tests(data_for_analysis, h)
})))

# Display results
print("QUESTION 4: STATISTICAL TEST COMPARISON")
print(all_comparison_results)

```

Conclusion:

Based on the Shapiro-Wilk test results from Question 2, all hormones are non-normally distributed in both groups. Therefore, the t-test is not appropriate as it assumes normality.

Between the Wilcoxon rank-sum test and Brunner-Munzel test, both are non-parametric tests suitable for non-normal data. Levene's test showed equal variances for all hormones, so Wilcoxon is sufficient. The Brunner-Munzel test is also valid and provides similar results.

Recommendation: Use Wilcoxon rank-sum test or Brunner-Munzel test since the data violates normality. Given equal variances across all hormones, Wilcoxon is the appropriate choice.

5. Constructing a correlation heatmap for all hormones separately in each of the two groups, selecting the correlation method (Pearson or Spearman) based on the normality of the data, and interpret the resulting patterns.

R code:

```
install.packages("corrplot")
library(corrplot)
# Determine correlation method based on normality
determine_method <- function(data, vars) {
  # Test normality for each variable
  normality <- sapply(vars, function(v) {
    if(sum(!is.na(data[[v]])) < 3) return(FALSE)
    test <- shapiro.test(data[[v]][!is.na(data[[v]])])
    return(test$p.value > 0.05)
  })

  # Count normal variables
  normal_count <- sum(normality, na.rm = TRUE)

  cat(sprintf("• %d out of %d hormones are normally distributed\n",
              normal_count, length(vars)))

  # If majority are normal, use Pearson; otherwise Spearman
  if(normal_count >= length(vars) * 0.7) {
    cat(" → Using PEARSON correlation (data is predominantly normal)\n")
    return("pearson")
  } else {
    cat(" → Using SPEARMAN correlation (data is predominantly non-
normal)\n")
    return("spearman")
  }
}
```

```
# Create correlation heatmaps for each group
pdf("Correlation_Heatmaps.pdf", width = 12, height = 6)

for(group_level in c("0", "1")) {
  # Subset data
  group_data <- data_for_analysis[data_for_analysis$outcome == group_level,
                                   hormone_vars]

  # Remove rows with any NAs
  group_data_clean <- na.omit(group_data)

  cat(sprintf("\nAnalyzing Group %s:\n", group_level))
  cat(sprintf("• Sample size after removing NAs: %d\n",
              nrow(group_data_clean)))

  # Determine correlation method
  cor_method <- determine_method(group_data_clean, hormone_vars)

  # Calculate correlation matrix
  cor_matrix <- cor(group_data_clean, method = cor_method)

  # Perform correlation significance test
  cor_test_matrix <- cor.mtest(group_data_clean, conf.level = 0.95)

  # Create heatmap with significance indicators
  par(mfrow = c(1, 1))

  corrplot(cor_matrix,
           method = "color",
           type = "upper",
```

```

        order = "hclust",
        addCoef.col = "black",
        number.cex = 0.7,
        tl.col = "black",
        tl.cex = 0.8,
        tl.srt = 45,
        p.mat = cor_test_matrix$p,
        sig.level = c(0.001, 0.01, 0.05),
        insig = "label_sig",
        pch.cex = 0.9,
        pch.col = "red",
        title = paste("Correlation Heatmap - Group", group_level,
                      "\n(Method:", cor_method, ")"),
        mar = c(0, 0, 3, 0))

# Add legend for significance
legend("bottom",
      legend = c("* p<0.05", "*** p<0.01", "**** p<0.001"),
      horiz = TRUE,
      bty = "n",
      cex = 0.8)
}

dev.off()

# Create comparative visualization
par(mfrow = c(1, 2))

for(group_level in c("0", "1")) {
  group_data <- data_for_analysis[data_for_analysis$outcome == group_level,

```

```

                                hormone_vars]

group_data_clean <- na.omit(group_data)
cor_method <- determine_method(group_data_clean, hormone_vars)
cor_matrix <- cor(group_data_clean, method = cor_method)

corrplot(cor_matrix,
          method = "circle",
          type = "upper",
          order = "hclust",
          tl.col = "black",
          tl.cex = 0.7,
          tl.srt = 45,
          title = paste("Group", group_level),
          mar = c(0, 0, 2, 0))
}

par(mfrow = c(1, 1))

```

Interpretation of Correlation Patterns

Overall pattern:

Both groups show predominantly weak to moderate correlations among hormones. No strong correlations were observed in either group.

Key findings in Group 0:

hormone3 and hormone4 show moderate positive correlation ($r = 0.587$). Other correlations are weak.

Key findings in Group 1:

hormone3 and hormone4 maintain moderate positive correlation ($r = 0.564$). hormone7 and hormone10_generated correlation increased from 0.190 to 0.269. hormone5 and hormone10_generated correlation increased from 0.011 to 0.216. hormone7 and hormone8 correlation increased from 0.088 to 0.136.

Interpretation:

The consistent moderate correlation between hormone3 and hormone4 in both groups suggests a stable biological relationship unaffected by outcome status. Group 1 shows slightly stronger inter-hormone correlations overall, particularly involving hormone5, hormone7, and hormone10_generated. This may indicate compensatory or pathological hormonal clustering associated with the outcome condition. However, all correlations remain weak to moderate, suggesting that these hormones largely function independently rather than as a tightly coordinated system in both groups.